

LABID AL NAHIYAN

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ABOUT ME

I am Labid Al Nahiyen, a recent graduate of Computer Science and Engineering from BUET, currently working as a Machine Learning Engineer (R&D). I am deeply passionate about Large Language Model (LLM) security, AI safety-alignment, and computer vision, driven by a goal to build robust, trustworthy, and scalable AI systems.

EDUCATION

Feb 2020 - Mar 2025 **Bangladesh University of Engineering and Technology**
BSc in Computer Science and Engineering

RESEARCH INTERESTS

Large Language Model (LLM) Security, AI-based Program Analysis, Software Engineering for AI, Computer Vision

RESEARCH EXPERIENCE

When Safety Blocks Sense: Measuring Semantic Confusion in LLM Refusals

Co-Author | [arXiv:2512.01037](#)

- Introduced "**Semantic Confusion**," a failure mode capturing local inconsistencies in LLM refusals where models reject harmless paraphrases of accepted intents.
- Developed **ParaGuard**, a 10k prompt corpus of controlled paraphrase clusters to evaluate model consistency.
- Proposed token-level metrics (Confusion Index, Rate, and Depth) using embeddings and perplexity signals to diagnose safety-alignment gaps in modern LLMs.

Digging into a New Contextually Appropriate Loss Function for Dense Crowd Tracking (Undergraduate Thesis)

Supervisor: Prof. A. B. M. Alim Al Islam

- Proposed a size-aware, gradient-stabilized loss function to optimize the tracking of small, overlapping objects in dense crowds.
- Integrated the function into **YOLO-based models** and evaluated on JHU-CROWD++ and SCUT-HEAD.
- Achieved superior mAP@50 and F1-scores compared to baseline loss functions including CloU, DloU, and GloU.

WORK EXPERIENCE

Aug 2025 - Present **TigerIT Bangladesh Ltd.** Dhaka, Bangladesh
Machine Learning Engineer (R&D)

- Reduced False Non-Match Rate (FNMR) by **41%** (from 0.0151 to 0.0089 @ FMR $\leq 10^{-2}$) for the **NIST MINEX III** benchmark by developing an optimized minutiae-based template generator.
- Engineered a **custom loss function** to address non-linear elastic distortions and rotational variances, significantly improving fingerprint matching accuracy and robustness.

Technologies: Python, PyTorch, C++, CUDA, Linux

May 2025 - July 2025 **Intellesphere.AI** Remote
Full-Stack Developer

- Built and structured responsive frontend components based on UI/UX designs.
- Integrated RESTful APIs for seamless frontend-backend communication and dynamic data rendering.
- Tested, debugged, and optimized the full functionality and performance of the live-deployed web application.

Technologies: React.js, Express.js, TypeScript

PROJECTS

SvelteKit, LLM, RAG **Shop Genie - AI E-commerce** Github
Developed an AI-powered shopping assistant enabling users to find products through natural language search. Scraped data for over 10,000 products across various websites for fast and robust product search.

SvelteKit, Tailwind, PostgreSQL	NerdHerd - Educational Platform Built an all-in-one student platform with classes, video calls, community features, and quizzes. Handled frontend design and integrated frontend-backend systems seamlessly.	Github
Arduino, Q-Learning	Bidirectional Autonomous Robot Built a bi-directional autonomous robot using Hierarchical Q-learning for real-time crawling and intelligent obstacle avoidance. Integrated live policy adaptation with a custom Arduino setup.	Website
C, Bison, Flex, YACC	Subset of C compiler Designed and implemented core components of a C compiler, including lexical analysis, parsing, symbol table management, and code generation.	Github
ReactJS, ExpressJS, Firebase	Book Shop Developed core functionalities allowing users to add books and place orders. Designed and implemented an admin dashboard for efficient monitoring and management of all book-related activities.	Github

TECHNICAL SKILLS

Languages:	C++, Python, Java, SQL, TypeScript
Frameworks:	PyTorch, React.js, Next.js, Svelte, Tailwind CSS, Express.js
Tools:	GitHub, Hugging Face, Docker, Linux, Bison, Flex
Database:	PostgreSQL, Firebase

AWARDS AND ACHIEVEMENTS

- Champion - OpenAPI Hackathon, 11th National ICT Fest 2024 (IUT).
- 7th Place - SEC Inter-University Junior Programming Contest 2022 (Team: BUET Fast & Furious).

REFERENCES

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 Professor
 Department of CSE, BUET.
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